

SHIVAM ARORA

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EDUCATION

Villanova University

May 2027

Bachelor of Science (Honors) in Computer Science & Economics

- **Relevant Coursework:** Data Structures & Algorithms, Computer Systems, Database Systems, Linear Algebra, Large-Language Model, Web & Mobile Development, Machine Learning , Agentic , Discrete Math

EXPERIENCE

Handshake AI

Feb 2026 – Present

AI Fellowship – Applied LLM Engineering

- **Technologies:** Python, TypeScript, LLM prompting, eval harnesses, Git, CI/CD
- Built prompt-eval pipelines with rubric scoring and regression sets to quantify code-assistant accuracy and failure patterns.
- Shipped PRs across production codebases—tests, refactors, and edge-case fixes—improving code quality and coverage.
- Defined a failure-mode taxonomy and review checklists adopted team-wide for structured regression tracking.

Esri

June 2025 – August 2025

Software Engineering Intern – AI Development & Integration

- **Technologies:** Python, FastAPI, Celery, ArcPy, REST APIs, Docker, Git, scikit-learn, spaCy, AWS
- Architected a telemetry and health-check service for AI-driven geospatial workflows, integrating model outputs with structured alerting to reduce unhandled pipeline errors by **40%**.
- Built FastAPI + Celery orchestration that translated NLP-classified intents into chained ArcPy pipelines of up to **10** tool executions, completing end-to-end in under **2s**.
- Added validation gates, retry policies, idempotency guards, and structured logging around model outputs to enforce safe, observable, and auditable production execution.

Villanova Research

Jan 2025 – Jan 2026

Software Engineer & Researcher – Real-Time Financial Analytics Platform

- **Technologies:** Python, FastAPI, asyncio, WebSockets, React, D3, Postgres, Redis, Git, CI/CD
- Built a multi-agent FastAPI backend for real-time financial signal processing with asyncio concurrency, serving parallel analytical agents with end-to-end latency under **50ms**.
- Implemented fault-tolerant queues, Redis caching, and backpressure controls with WebSocket streaming to a React + D3 dashboard; validated **99.9%** uptime under sustained load testing.
- Enforced CI-tested interface contracts and strict typing across services, cutting integration errors by **30%**.

Ernst & Young (EY)

June 2024 – August 2024

Technical Solutions Intern

- **Technologies:** SQL, Excel, data validation scripts, JIRA, Visio, SAP S/4HANA
- Mapped finance, procurement, and supply-chain flows in SAP S/4HANA, surfacing **5** critical process and integration gaps.
- Built SQL validation checks and batch jobs, cutting integration errors by **15%** and improving processing throughput by **20%**.
- Led workshops for **20+** users, driving **80%** dashboard adoption and improving master-data accuracy by **15%**.

TECHNICAL SKILLS

Languages: C++, Java , Python , JavaScript, HTML, CSS, SQL

Frameworks & Libraries: Node.js, Express.js, React, Vue, Tailwind, REST API, Bootstrap, FastAPI, NLTK, AngularJS, LangGraph. LlamaIndex, CrewAI, Scikit-learn, PyTorch, TensorFlow, Ollama

Developer Tools: Git, GitHub, Postman, Jenkins, Jupyter, Docker, AWS, Google Cloud, Kubernetes

PROJECTS

VibeVisor | *Electron, React, TypeScript, Python, WebSockets, IPC, AST parsing, Git hooks*

- Built an Electron overlay that monitors AI coding agents (Claude Code, Cursor) in real time, capturing diffs, terminal output, and file changes via IPC listeners and Git hooks to track full session history and flag regressions before commit.
- Implemented automated **SDLC gate checks** system design validation, test coverage analysis, documentation completeness scoring, and **deploy-readiness auditing surfacing actionable warnings** in-overlay to reduce manual review cycles.

DataStackBuilder | *Python, FastAPI, React, TypeScript, Vite, SQLite, Postgres, Redis, Docker*

- Built an open-source data pipeline platform with a **visual DAG design canvas**, real-time simulation engine for latency/throughput/cost analysis, and a production control plane supporting versioning, backfills, **CDC connectors**, and lineage tracking.
- Designed a **pipeline economics reviewer** that imports dbt metadata, merges observed topology from OpenLineage events, and runs baseline-vs-proposed cost/latency diffs with **CI gate policy evaluation (PASS/WARN/FAIL)** and Markdown report generation.

LEADERSHIP & ACTIVITIES

Resident Assistant RA (425+ residents); Classroom Support Tech (Level 2); VU Hospitality Associate; VISO M&O Chair; SASE member; Club Badminton, VU Programming ; Selected AI Mentor for AI Presidential Challenge